

In Section 2, we give motivation from non-linear stochastic filtering, volatility modelling in finance, pathwise stochastic optimal control and mean-fields SDE with common noise system; topics also closely related to [11, 12]. We do not strive for precise statements, but try to explain how rough stochastic processes naturally emerge, together with motivation for what is to come: Section 3 deals with the mathematics of randomisation of rough Itô processes, one obstacle being the potentially uncountable number of rough paths that figure as parameter in (1.1). The remaining section discuss applications to RSDE and then also mean-field RSDEs. The body of work surrounding RSDEs, control and common noise also resonates with early remarks made by the authors of [10], conjecturing the use of rough paths for conditioned forward stochastic dynamics (in their context of mean field games with common noise).

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2 Motivating examples

2.1 Non-linear stochastic filtering

We follow [16, 8]. One has an observation process $Y_t = \int_0^t h(X_s, Y_s) ds + B_t^\perp$ where the signal process X is assumed to have “correlated” dynamics,

$$\begin{aligned} dX_t &= \tilde{b}(t, X_t, Y_t) dt + \sigma(t, X_t, Y_t) dB_t(\omega) + f(t, X_t, Y_t) dB_t^\perp(\omega) \\ &= b(t, X_t, Y_t) dt + \sigma(t, X_t, Y_t) dB_t(\omega) + f(t, X_t, Y_t) dY_t(\omega). \end{aligned} \quad (2.1)$$

Here, $(B, B^\perp)$ and (B, Y) are independent Brownians under some original measure $\mathbb{P}^o$ and $\mathbb{P}$, respectively, related by the Girsanov formula

$$\left. \frac{d\mathbb{P}^o}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \exp \left(\int_0^t h(s, X_s, Y_s) dY_s - \frac{1}{2} \int_0^t |h(s, X_s, Y_s)|^2 ds \right) =: \exp(I_t).$$

Formally at least, the conditional dynamics of X_t given the observation $\{Y_t : 0 \leq t \leq T\}$ should be captured by the solution $X^\mathbf{Y}$ to a rough SDE of the form (1.5)²

$$dX_t^\mathbf{Y} = b(t, X_t^\mathbf{Y}, Y_t) dt + \sigma(t, X_t^\mathbf{Y}, Y_t) dB_t(\omega) + (f, D_y f)(t, X_t^\mathbf{Y}, Y_t) d\mathbf{Y}_t, \quad (2.2)$$

together with the “rough stochastic” Girsanov exponent

$$I_t^\mathbf{Y} = \int_0^t (h, (D_x h f + D_y h))(s, X_s^\mathbf{Y}, Y_s) d\mathbf{Y}_s - \frac{1}{2} \int_0^t |h(s, X_s^\mathbf{Y}, Y_s)|^2 ds.$$

Bearing exponential integrability, one defines a flow of measures

$$\langle \mu_t^\mathbf{Y}, \varphi \rangle := \mathbb{E}[\varphi(X_t^\mathbf{Y}) \exp(I_t^\mathbf{Y})],$$

² Assuming regular t dependence in f .